

# QUIZ 2 REVIEW (CH1&2)

Shun/11/25 (shun4midx)

## OLD QUIZ 1

$$C_1 = \begin{cases} 00 = N \\ 01 = W \\ 10 = E \\ 11 = S \end{cases}, C_2 = \begin{cases} 000 \\ 011 \\ 101 \\ 110 \end{cases}, C_3 = \begin{cases} 00000 \\ 01101 \\ 10110 \\ 11011 \end{cases}$$

1) Hamming Distance

a)  $d(00101, 11011) = 4$

$$\begin{array}{c} 00101 \\ 11011 \\ \hline 0122 \end{array}$$

b)  $d(0122, 1020) = 3$

$$\begin{array}{c} 0122 \\ 1020 \\ \hline 0120 \end{array}$$

2) Hamming distance is a legitimate distance function:

(i)  $d(x, y) = 0$  if and only if  $x = y$

(ii)  $d(x, y) = d(y, x)$  for all  $x, y \in (F_q)^n$

(iii)  $d(x, y) \leq d(x, z) + d(y, z)$  for all  $x, y, z \in (F_q)^n$

3) Minimum distance:  $d(C_1) = 1$ ,  $d(C_2) = 2$ ,  $d(C_3) = 3$

4) Sufficient conditions for detection and correction

(i) A code  $C$  can detect up to  $s$  errors in any codeword if  $d(C) = s+1$ . (Fill in the minimum number)

(ii) A code  $C$  can correct up to  $t$  errors in any codeword if  $d(C) = 2t+1$ . (Fill in the minimum number)

## OLD QUIZ 2

1) Suppose  $d$  is odd. Show that a binary  $(n, M, d)$ -code exists if and only if a binary  $(n+1, M, d+1)$ -code exists.

" $\Rightarrow$ ": Consider any binary  $(n, M, d)$ -code we call  $C$ .

For each codeword  $x$  in  $C$ , append a bit at the end. Denote  $\hat{x} := \begin{cases} x_1 \dots x_n 0, & w(x) \text{ even} \\ x_1 \dots x_n 1, & w(x) \text{ odd} \end{cases}$ , where  $x = x_1 \dots x_n$

The new code is a binary  $(n+1, M, d+1)$ -code:

- $n+1$ : Each codeword now has  $n+1$  bits.
- $M$ : There are still  $M$  codewords
- $d+1$ : Take  $x \neq y \in C$ , where  $d(x, y) =: r \geq d$ .

$$\text{Then, } w(x) = w(y) \Leftrightarrow d(x, y) = w(x) + w(y) - 2w(x \cap y) \text{ is even}$$

$$\therefore d(\hat{x}, \hat{y}) = \begin{cases} r, & r \text{ even} \\ r+1, & r \text{ odd} \end{cases}$$

$$\text{When } r \text{ even, } d(\hat{x}, \hat{y}) \geq d, \text{ but } d \text{ is odd, so } d(\hat{x}, \hat{y}) \geq d+1 \Rightarrow d(\hat{x}, \hat{y}) \geq d+1$$

$$\text{When } r \text{ odd, } d(\hat{x}, \hat{y}) \geq d+1$$

" $\Leftarrow$ ": Consider any binary  $(n+1, M, d+1)$ -code  $C$ .

There must exist a position where not all  $M$  codewords share the same bit. Let's call that position  $i$ .

$\forall x = (x_1, \dots, x_{n+1}) \in C$ , we define  $\hat{x} := (x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{n+1})$

The set of  $\{\hat{x} \mid x \in C\} =: C'$  is an  $(n, M, d)$ -code

- $n$ : Each codeword's length decreased by 1
- $M$ : Assume not, then  $\exists x \neq y \in C$ , s.t.  $\hat{x} = \hat{y}$ . Then,  $d(x, y) = 1$ , but  $d(x, y) \geq d+1 \geq 2$  since  $d$  is odd. \*
- $d$ :  $\forall x \neq y \in C$ , we know  $d(x, y) \geq d+1$ .

$$d(\hat{x}, \hat{y}) = \begin{cases} d(x, y) & , x_i = y_i \\ d(x, y) - 1 & , x_i \neq y_i \end{cases} \geq (d+1) - 1 = d \quad \square$$

2) Two  $q$ -ary codes are called equivalent if one can be obtained from the other by a combination of operations of the following types:

- Permutation of positions of a code.
- Permutation of symbols of a position.

3) Show that any  $q$ -ary  $(n, M, d)$ -code over an alphabet  $\{0, 1, 2, \dots, q-1\}$  is equivalent to an  $(n, M, d)$  code which contains the all-zero vector  $0 = 00 \dots 0$ . [Shun/1135 \(@shun4mid\)](#)

Suppose the original code had  $x = (x_1, \dots, x_n)$ .

For every position  $i$  from 1 to  $n$ , do a permutation of symbols swapping  $x_i$  and 0.

Then,  $x$  becomes  $(0, \dots, 0)$ , while still being an equivalent  $(n, M, d)$ -code.  $\square$